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Batch 15

Algorithm:

1. **Start.**
2. Declare two character arrays `s` and `t`.
3. Declare an integer array `freq[26]` and initialize all its elements to `0`.
4. Read strings `s` and `t` from the user.
5. Pass the addresses of `s` and `t` to the function `check_anagram()`.
6. Inside `check_anagram()`:
   - Traverse string `s` using a pointer.
   - For each character in `s`, increase its corresponding frequency in `freq[]` by `1`.
   - Traverse string `t` using a pointer.
   - For each character in `t`, decrease its corresponding frequency in `freq[]` by `1`.
7. Traverse all 26 elements of `freq[]`.
8. If **any frequency is not equal to `0`**, print **`Not Anagram`** and stop the function.
9. If **all frequencies are `0`**, print **`Anagram`**.
10. **Stop.**

test table

Test Case Table – Anagram Program

Test Case	String `s`	String `t`	Expected Output	Purpose
1	`listen`	`silent`	Anagram	Same characters with same frequencies
2	`hello`	`world`	Not Anagram	Different characters
3	`anagram`	`nagaram`	Anagram	Same characters in different order
4	`rat`	`car`	Not Anagram	Same length but different characters
5	`abc`	`abcd`	Not Anagram	Strings have different lengths
6	`aabbcc`	`abcabc`	Anagram	Repeated characters with same frequencies
7	`aabb`	`abbb`	Not Anagram	Same characters but different frequencies
8	`a`	`a`	Anagram	Single-character boundary case
9	`a`	`b`	Not Anagram	Different single characters

10	abc	cba	Anagram	Completely reversed order

C program

```
#include <stdio.h>

void check_anagram(char *s, char *t)
{
    int freq[26] = {0};
    int i = 0;

    // Count characters in string s
    while (*(s + i) != '\0')
    {
        freq[*(s + i) - 'a']++;
        i++;
    }

    // Subtract characters in string t
    i = 0;
    while (*(t + i) != '\0')
    {
        freq[*(t + i) - 'a']--;
        i++;
    }

    // Check all character frequencies
    for (i = 0; i < 26; i++)
    {
        if (freq[i] != 0)
        {
            printf("Not Anagram\n");
            return;
        }
    }

    printf("Anagram\n");
}

int main()
{
    char s[100], t[100];

    printf("String s: ");
    scanf("%99s", s);

    printf("String t: ");
    scanf("%99s", t);

    check_anagram(s, t);

    return 0;
}
```

Execution

```
Last login: Thu Oct 29 08:14:41 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano day60.c
bhavya@Bhavyas-MacBook-Air ~ % clang day60.c -o day60
bhavya@Bhavyas-MacBook-Air ~ % ./day60
String s: listen
String t: silent
Anagram
bhavya@Bhavyas-MacBook-Air ~ % ./day60
String s: hello
String t: world
Not Anagram
bhavya@Bhavyas-MacBook-Air ~ %
```